

Fundamentals of Machine Learning

Tutorial 1: From Learning Problems to Probabilistic Classification

Central question

How do representation, loss, finite data, and modeling assumptions determine what a learning system predicts and how reliably it generalizes?

Objectives

By the end of the tutorial, students should be able to

- formulate a learning problem from an application;
- derive decisions from probabilities and costs;
- connect representation, model capacity, and generalization;
- distinguish training, validation, and test information;
- compare discriminative and generative classification.

Problems

1. From a security objective to a learning problem

15 minutes

A security operations center receives network flows. For each completed flow, the raw record contains packet times, addresses, ports, protocol information, and byte counts. A ground-truthing process assigns each flow a benign or attack label using incident information or known attack-scenario metadata. An attack label may indicate membership in an attack episode, even when the individual flow resembles benign traffic. The deployed system must score a new flow and decide whether to raise an alert.

- (a) Identify the physical object, the representation $\mathbf{x}$, the target Y , the model output, and the final action.
- (b) Suppose some attack-labelled and benign flows have the same, or very similar, observed features. Explain why a classifier using only these features may be unable to separate them perfectly. Does this necessarily mean that the labels are incorrect?
- (c) Explain why a posterior estimate $p_{\hat{\theta}}(y = 1 \mid \mathbf{x})$ and the alert decision are not the same object.
- (d) Give one feature that is available at decision time and one feature that would create target leakage.
- (e) Suppose most attacks in the collected data occurred at night, while most benign flows occurred during the day. Explain how time of day could become a learning shortcut. Why is this problem different from target leakage?

- (f) Briefly describe one cybersecurity task for each of the following paradigms: supervised learning, self-supervised learning, and reinforcement learning. For each task, identify the information that provides the learning signal.

2. The optimal action depends on the loss

10 minutes

For a flow with features $\mathbf{x}$, let

$$p = p(y = 1 \mid \mathbf{x}),$$

where class 1 denotes an attack. A false alert costs $C_{\text{FP}} = 2$ and a missed attack costs $C_{\text{FN}} = 8$. Correct decisions have zero cost.

- (a) Derive the conditional risk of declaring attack and of declaring benign.¹ Derive the optimal probability threshold.
- (b) Apply the rule to three flows with posterior probabilities

$$0.08, \quad 0.35, \quad 0.85.$$

- (c) Human review is now available. Assume that requesting review incurs a fixed loss of 0.9, regardless of the true class, and that review then resolves the case correctly. Derive the probability regions for declaring benign, requesting review, and declaring attack.
- (d) Explain why changing the decision threshold to reflect error costs is different from correcting a model for class imbalance.

3. KNN inherits the geometry of the representation

15 minutes

A query flow has duration 50 ms and 10 packets:

$$\mathbf{x} = (50, 10).$$

Two labeled flows are

$$\mathbf{x}_A = (60, 100), \quad \mathbf{x}_B = (400, 12).$$

- (a) Using raw Euclidean distance, determine the nearest flow.
- (b) Scale duration by 100 ms and packet count by 20 packets. Determine the nearest flow after scaling.
- (c) Let $\mathbf{z}$ denote the scaled features from part (b). Suppose flow A is labeled attack and flow B is labeled benign. For 1-NN with ordinary Euclidean distance in the scaled (z_1, z_2) plane, derive and plot the decision boundary between the two training flows. Mark the query flow and identify its predicted class.
- (d) To control the relative importance of the two features, consider the weighted squared distance

$$d_w^2(\mathbf{z}, \mathbf{z}') = w(z_1 - z'_1)^2 + (z_2 - z'_2)^2, \quad w > 0.$$

Derive the critical value of w at which the nearest flow changes. Which flow is nearest on each side of this value? Interpret the result, and explain why w should be selected using validation data.

4. Audit a generalization claim

15 minutes

A team collects 200,000 network-flow records. One attack episode can generate many nearly duplicate flows. The team performs the following experiment:

¹The conditional risk of an action is the average loss expected if that action is taken for a flow with given features.

- standardize all records before splitting;
 - randomly split individual rows into training and test sets;
 - try 100 model and feature configurations;
 - select the configuration with the highest test accuracy;
 - report that same accuracy as final performance.
- (a) Identify every mechanism that can make the reported performance too optimistic.
- (b) Propose a training, validation, and test protocol for each goal:
- new flows drawn from the same simulated data-generating process;
 - later traffic when user behavior, services, or attack patterns may change over time;
 - new attack episodes that differ from those represented during training.
- (c) A test set contains 10,000 flows, including 100 attacks. A classifier produces

$$TP = 70, \quad FP = 180, \quad FN = 30, \quad TN = 9720.$$

Compute accuracy, precision, recall, and F_1 .

- (d) Compare this classifier with the rule that predicts every flow as benign. Why is accuracy insufficient here?
- (e) State where scaling, missing-value treatment, and feature selection must be fitted in a cross-validation pipeline.

5. Capacity, regularization, and validation

15 minutes

Three nested model families are fitted to the same training sample:

	$\mathcal{H}_1$	$\mathcal{H}_2$	$\mathcal{H}_3$
training error	0.20	0.08	0.00
validation error	0.22	0.11	0.19

- (a) Explain why the minimum training error cannot increase from $\mathcal{H}_1$ to $\mathcal{H}_3$.
- (b) Which family would you select? Diagnose the other two using underfitting, estimation variance, and excessive capacity where appropriate.
- (c) After collecting much more training data, the validation error of $\mathcal{H}_3$ decreases to 0.10. Explain why capacity cannot be judged without reference to sample size.
- (d) Two model configurations have the same population error $R = 0.20$. Their validation errors are

$$\hat{R}_1 = R + \varepsilon_1, \quad \hat{R}_2 = R + \varepsilon_2,$$

where ε_1 and ε_2 are independent $\mathcal{N}(0, \sigma^2)$ variables with $\sigma = 0.02$. The configuration with the smaller validation error is selected. Using

$$\min(a, b) = \frac{a + b - |a - b|}{2}, \quad \mathbb{E}|\varepsilon_1 - \varepsilon_2| = \frac{2\sigma}{\sqrt{\pi}},$$

derive the expected reported validation error. Compare it with the population error and interpret the difference.

6. Discriminative probabilities, decisions, and prior shift

20 minutes

A binary discriminative model produces score $s_{\hat{\theta}}(\mathbf{x})$ and posterior estimate

$$r_t(\mathbf{x}) = p_{\hat{\theta}}(y = 1 \mid \mathbf{x}) = \frac{1}{1 + e^{-s_{\hat{\theta}}(\mathbf{x})}}.$$

- (a) For $s_{\hat{\theta}}(\mathbf{x}) = 1.2$, compute the posterior estimate. If the true label is 1, compute the cross-entropy loss.
- (b) Explain why minimizing empirical cross-entropy is equivalent to maximizing the conditional likelihood of the observed labels under an i.i.d. assumption.
- (c) Let $C_{\text{FP}} = 4$ and $C_{\text{FN}} = 16$. Determine the cost-sensitive threshold and the decision for this flow.
- (d) Consider a two-feature logistic model with

$$s_{\hat{\theta}}(\mathbf{x}) = -3 + 2x_1 + x_2.$$

Plot its decision boundary first with threshold 0.5, then with the cost-sensitive threshold found in part (c). On each plot, indicate the region classified as attack. Explain the movement of the boundary.

- (e) The training attack prior is $\pi_t = 0.20$, while the deployment prior is $\pi_d = 0.05$. Assume prior shift and a calibrated training posterior $r_t(\mathbf{x}) = 0.60$. Compute

$$r_d(\mathbf{x}) = \frac{(\pi_d/\pi_t)r_t(\mathbf{x})}{(\pi_d/\pi_t)r_t(\mathbf{x}) + \frac{1-\pi_d}{1-\pi_t}[1-r_t(\mathbf{x})]}.$$

- (f) Distinguish the prior-shift correction above from choosing a decision threshold based on operational costs.
- (g) When the minority-class sample is very small, explain why weighting or resampling may help estimation even when no prior shift is present.

7. Generative classification: assumptions become visible

20 minutes

Part I: Gaussian class models. Let $\mathbf{x} = (x_1, x_2)^\top$ contain two standardized numerical features, such as log-duration and log-packet count. Suppose both classes have the same correlated covariance:

$$p_{\theta}(\mathbf{x} \mid y = 0) = \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}), \quad p_{\theta}(\mathbf{x} \mid y = 1) = \mathcal{N}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}),$$

where

$$\boldsymbol{\mu}_0 = (0, 0)^\top, \quad \boldsymbol{\mu}_1 = (2, 2)^\top, \quad \boldsymbol{\Sigma} = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}, \quad |\rho| < 1,$$

with priors $\pi_0 = 0.75$ and $\pi_1 = 0.25$.

- (a) Show that

$$\boldsymbol{\Sigma}^{-1} = \frac{1}{1-\rho^2} \begin{pmatrix} 1 & -\rho \\ -\rho & 1 \end{pmatrix}.$$

Starting from the Gaussian log posterior, remove terms common to all classes and derive the LDA score

$$s_c(\mathbf{x}) = \mathbf{x}^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_c - \frac{1}{2} \boldsymbol{\mu}_c^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_c + \log \pi_c.$$

Explain why comparing these scores produces a linear boundary.

- (b) Set $\rho = 0.5$. Derive explicit expressions for $s_0(\mathbf{x})$ and $s_1(\mathbf{x})$, then write the decision boundary $s_0(\mathbf{x}) = s_1(\mathbf{x})$ as an equation involving $x_1 + x_2$. Plot the boundary and mark the two class means.
- (c) For $\mathbf{x} = (1.25, 1.25)^\top$, determine the predicted class and compute the posterior probability of class 1 by normalizing $e^{s_0(\mathbf{x})}$ and $e^{s_1(\mathbf{x})}$.

- (d) Repeat parts (b) and (c) with equal class priors. Explain how the prior moves the LDA boundary while the class-conditional models remain unchanged. Add the equal-prior boundary to the plot from part (b), and mark the test point $\mathbf{x} = (1.25, 1.25)^\top$.

Part II: Naive Bayes. Let $x_1 = 1$ indicate destination port 22, and let $x_2 = 1$ indicate that flow duration is below a predefined threshold τ .² Suppose

	$p(x_1 = 1 \mid y = c)$	$p(x_2 = 1 \mid y = c)$	π_c
attack	0.60	0.70	0.10
benign	0.05	0.20	0.90

- Under the Naive Bayes assumption, compute the two unnormalized class scores for a flow with $x_1 = x_2 = 1$. Compute the attack posterior.
- State the conditional-independence assumption used in the calculation. Why might it fail for network-flow features?
- Explain how correlated features can repeat the same evidence and produce overconfident probabilities.
- A port value appears in validation but never appeared in training. Describe how an explicit “other” category and additive smoothing prevent a zero likelihood.
- Give one reason to prefer the discriminative route and one reason to prefer the generative route for this application.

8. Detecting an attack from a sequence of flows

20 minutes

A detector observes completed flows from one source host. The first flow is benign background traffic, and a port-scan episode begins at $a_2 = 17$ seconds. Each row gives the flow start time a_i , completion time t_i , destination port, connection outcome, and flow-level attack score. The score s_i becomes available at t_i and is produced by a previously fitted classifier using the full feature vector of flow i ; the table shows only the fields needed for this exercise.

i	1	2	3	4	5	6
a_i (seconds)	0	17	30	43	56	73
t_i (seconds)	2	18	31	44	57	74
destination port	80	22	23	25	445	3389
outcome	success	fail	fail	fail	fail	fail
s_i	0.10	0.18	0.22	0.24	0.28	0.26

The flow-level detector raises an alert when $s_i \geq 0.70$. Whenever flow i is completed at time t_i , a second detector forms a 60-second window ending at t_i . The window contains flow i and every earlier flow from the same source that was completed after $t_i - 60$. The detector recomputes its window features and raises an alert when the window contains at least four flows, at least four distinct destination ports, a failure fraction of at least 0.75, and a flow-score sum of at least 0.70. This detector combines accumulated model evidence with features describing recent behavior. Part (f) considers score accumulation without the window features. The fixed rule makes the calculations visible. In practice, a second supervised model can learn a window-level attack score from these features, using windows constructed from the training episodes.

- Which flows trigger the flow-level detector? What does this show about classifying each flow separately?
- Compute the four window features after each flow until the first alert. At what time is the scan first detected? Give the delay from the beginning of the episode and the delay from the completion of its first flow.

²The threshold is fixed using domain knowledge or training and validation data.

- (c) Repeat the detection procedure using a 20-second window. Determine whether an alert is raised, then explain the benefit and limitation of shortening the window.
- (d) For the decisions made in part (c), state which information is available at each decision time. Give two pieces of information that would introduce future-information leakage.
- (e) Suppose the source IP is a public NAT address shared by several devices. Explain how this can affect the window features. Suggest a better grouping key when suitable network information is available.
- (f) Consider discounted score accumulation

$$A_i = 0.5A_{i-1} + s_i, \quad A_0 = 0.$$

Compute A_i until it first exceeds an alert threshold of 0.45. Compare its detection time with the moving-window detector and identify one difficulty in choosing the accumulation rule.

- (g) Describe a leakage-safe train, validation, and test split when several windows are generated from the same attack episode.